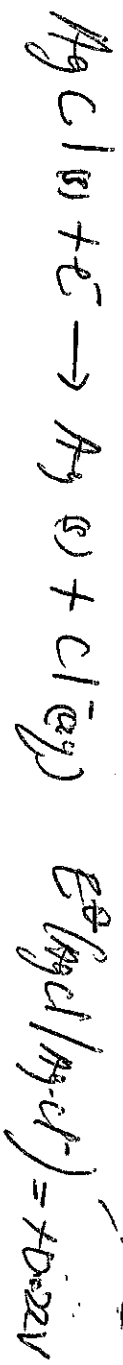
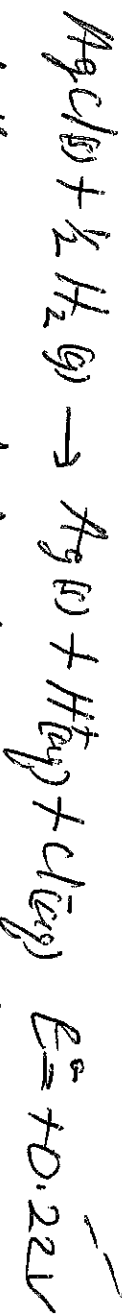


Although a standard potential is written as though it refers to a half-reaction such as;



It should be understood that these equations are only shorthand for writing;



and that the standard potential is determined by properties of the hydrogen electrode as well as the couple to which the potential refers. Table-1 lists standard potentials at 298K.

Table 1 Standard Potentials at 298K.

<u>Couple</u>	<u>$E^\ominus/\text{V}$</u>
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.61
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$	+0.34
$\text{AgCl(s)} + \text{e}^- \rightarrow \text{Ag(s)} + \text{Cl}^-(\text{aq})$	+0.22
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)}$	-0.76
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na(s)}$	-2.71

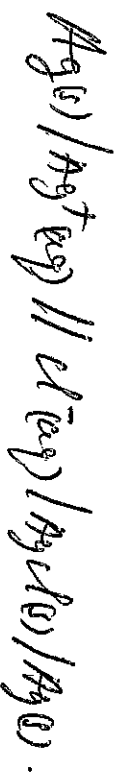
An important feature of standard emf of cells and standard potentials of electrode ~~is well as~~ that they are unchanged if the chemical equation for the cell reaction or a half-reaction is multiplied by a numerical factor. A numerical factor increases the value of the standard Gibbs energy for the reaction. Hence, it is also increased number of e^- transferred by the same factor, and by equation

$$E^\ominus = \frac{\Delta_r G^\ominus}{nF}$$

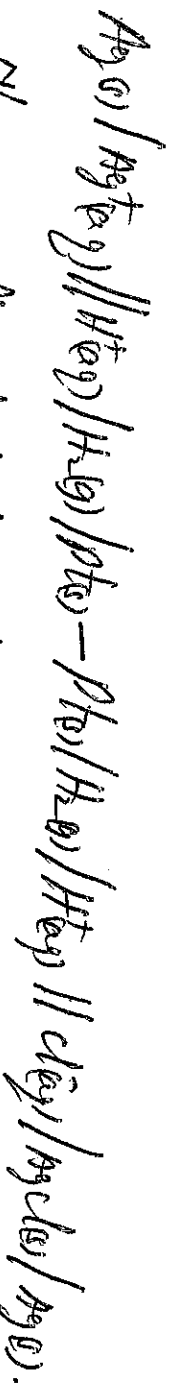
the value of $E^\ominus$ remains unchanged.

The cell emf in terms of individual Standard Potentials

The standard emf of a cell formed from any two electrodes is calculated by taking the difference of their standard potentials. This rule follows the fact that a cell such as



is equivalent to two cells joined back to back.



The overall potential of this composite cell, and therefore of the cell of interest, is

$$E^\ominus = E^\ominus(\text{AgCl} / \text{Ag, Cl}^-) - E^\ominus(\text{Ag}^+ / \text{Ag}) = -0.58 \text{ V}$$

The standard potentials in Table-1 can all be used in the same way, and the standard emf of the cell is obtained by subtracting the standard potential of the left-hand electrode from the standard potential of the right-hand electrode.

$$E^{\circ} = E^{\circ}(\text{right}) - E^{\circ}(\text{left})$$

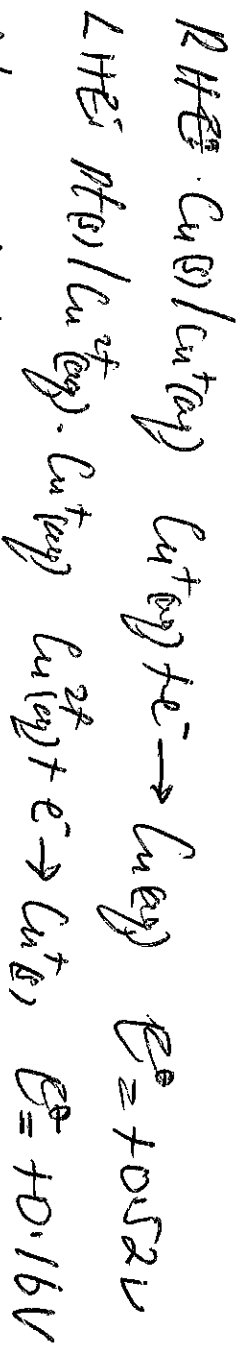
Because $\Delta G^{\circ} = -nFE^{\circ}$, it then follows that, if the result gives $E^{\circ} > 0$, then the corresponding cell reaction has $K > 1$.

Calculation:

To calculate the equilibrium constant for the disproportionation

$$2\text{Cu}^{+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \text{ at } 298.15\text{K}.$$

Solution:



The standard emf of the cell is therefore:

$$E^{\circ} = +0.52\text{V} - 0.16\text{V} = +0.36\text{V}$$

Then because $n=1$

$$\ln K = \frac{0.36\text{V}}{0.025693\text{V}} = \underline{\underline{14.06}}$$

$$K = \underline{\underline{1.2 \times 10^6}}$$

(from $\ln K = \frac{nFE^{\circ}}{RT}$... cell at eqn. from Nernst.)

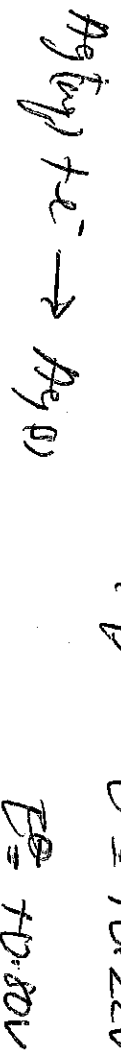
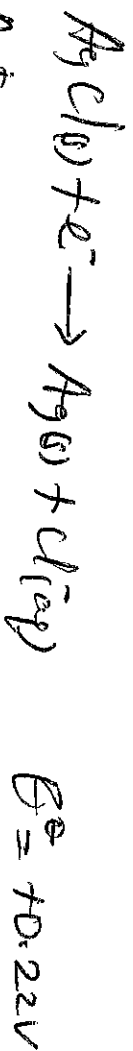
Example:

(2)

Estimate the solubility constant of silver chloride / the equilibrium constant for the dissolution of AgCl(s) and its solubility from cell potential data at 298-15K.

Solution

The dissolution can be expressed as the difference of the following half-reactions:



The standard emf is therefore -0.58V . Then, because $v=1$,

$$\ln K = \frac{vE^\ominus}{RT/F} = \frac{1 \times (-0.58\text{V})}{2.5693 \times 10^{-2}\text{V}} = \frac{-0.58}{2.5693 \times 10^{-2}} = -22.5742$$

$$K = \frac{1.6 \times 10^{-10}}{1}, \text{ because } a_{\text{Ag}}, a_{\text{Cl}} = 1, \text{ it follows}$$

$$\text{that, } S = 1.3 \times 10^{-5} \text{ mol kg}^{-1}$$

Measurement of Standard Potentials

The procedure for measuring a standard potential can be illustrated by constructing a precise cell, the silver chloride electrode - the measurement is made on the 'Harned cell'.



for which the Nernst equation is

$$E = E^{\circ}(AgCl/AgCl^{-}) - \frac{RT}{F} \ln \frac{a_{H_2O}}{(f_{H_2O}/P^{\circ})^{1/2}}$$

We shall set $f = P^{\circ}$ from now on, and for simplicity with the standard potential as E° , the activity can be expressed in terms of the molality b and the mean activity coefficient $\gamma_{\pm}$, through equation $\gamma_{\pm} = (\gamma_+ \gamma_-)^{1/2}$.

$$E = E^{\circ} - \frac{RT}{F} \ln b^2 - \frac{RT}{F} \ln \gamma_{\pm}^2$$

This expression rearranged to

$$E + \frac{2RT}{F} \ln b = E^{\circ} - \frac{2RT}{F} \ln \gamma_{\pm}$$

from the Debye-Huckel limiting law for a 1:1 electrolyte, we know that $\ln \gamma_{\pm} \propto -b^2$ (the natural log. used here is proportional to the common log. that appears

$$\log \gamma_{\pm} = -1/2 \cdot 2.303 / A b^{1/2} \quad \text{--- Debye-Huckel limiting law}$$

Therefore, with the constant of proportionality in this relation written as $(F/2RT)C$,
 where $A = 0.509$, 1 , ion strength I solution.

$$E + \frac{2RT}{F} \ln b = E^{\circ} + C b^{1/2}$$

The expression on the left is evaluated at a range of molalities, plotted against $b^{1/2}$ and extrapolated to $b=0$. The intercept at $b^{1/2}=0$ is the value of E° for the Silver/Silver chloride electrode.

Thermodynamic Functions

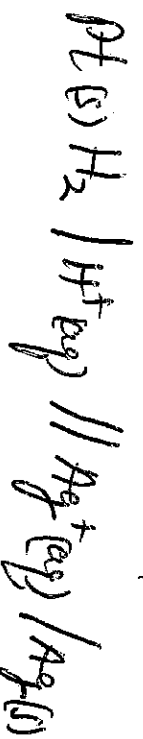
(25)

The standard emf of a cell is related to the standard reactions Gibbs energy through the equation

$$E^{\ominus} = \frac{-\Delta_r G^{\ominus}}{nF} \quad (\text{or } \Delta_r G^{\ominus} = -nFE^{\ominus}). \quad \text{Therefore, by measuring } E^{\ominus} \text{ we can obtain this important thermodynamic quantity. Its value can be used to calculate the Gibbs energy of formation of ions.}$$

Illustration

The cell reaction taking place in



$$E^{\ominus} = +0.7996 \text{ V}$$



$$\Delta_r G_m^{\ominus} = -\Delta_r G^{\ominus} / \text{Ag}^+(\text{aq})$$

Therefore, with $n=1$ we find

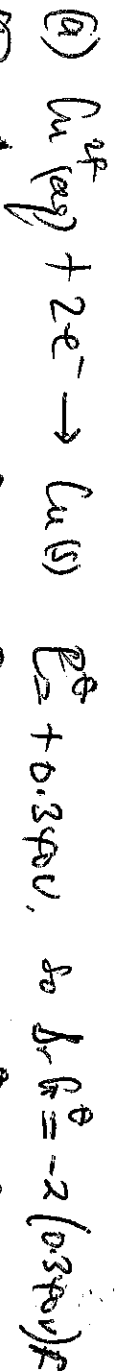
$$\Delta_r G_m^{\ominus}(\text{Ag}^+, \text{aq}) = -(-FE^{\ominus}) = +77.15 \text{ kJ mol}^{-1}$$

is in close agreement with the value in theory.

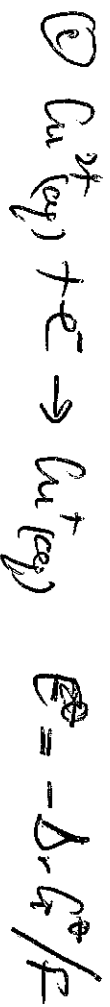
Example: Evaluating a Standard Potential from two others.
Given that the standard potentials of the Cu^{2+}/Cu and Cu^+/ Cu couples are $+0.340 \text{ V}$ and $+0.522 \text{ V}$, respectively, calculate $E^{\ominus}(\text{Cu}^{2+}, \text{Cu}^+)$.

Solution:

The electrode reactions are:-



The required reaction is:-



$\therefore$ because (c) = (a) - (b), the standard Gibbs energy of reaction

(c) is $\Delta_r G^\ominus = \Delta_r G^\ominus(\text{a}) - \Delta_r G^\ominus(\text{b}) = (-0.158\text{V}) \times F$

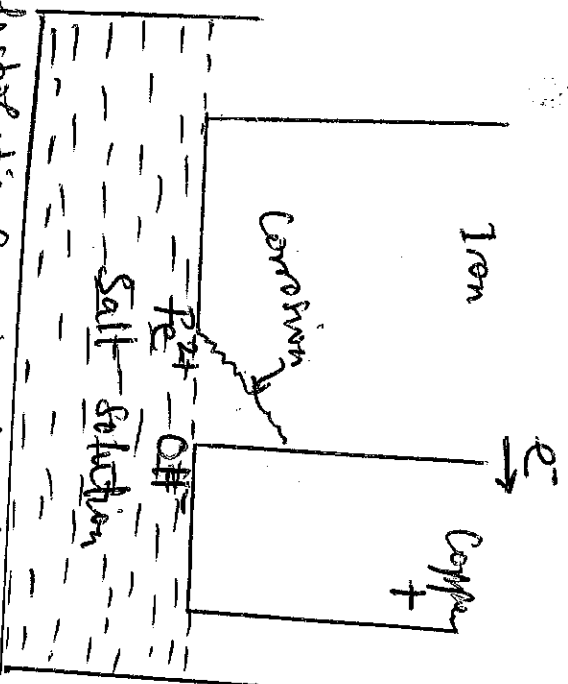
therefore, $E^\ominus = +0.158\text{V}$.

CORROSION OF METALS

Corrosion is a chemical reaction occurring at the surface of metals by which the metals are converted into chemical compounds. For example, iron in moist atmosphere turns into iron oxide and the phenomenon is known as the rusting of iron. Certain metals like gold, silver, platinum, etc, do not undergo these changes and are called 'noble' metals. Due to the slow metal oxidation and the formation of impermeable skin, the corrosion is not a rapid process, though the process is rapid in moist air, presence of sulphur dioxide and water.

As the corrosion is an electro-chemical process, a potential difference is set up between the parts of the surface, resulting in the formation of chemical cells.

If we have copper and iron junction dipped into a salt solution, copper, being more electropositive, will act as the positive pole & anode whereas iron will behave like a cathode. This will result in the flow of e^- from iron to copper. Inside the solution, iron is anode and copper is cathode as shown in the figure below.



Electrolytic Corrosion at the junction b/w Fe-Cu.

Due to anode reactions iron will be dissolved as ferrous ions, but at the cathode, Na^+ or H^+ ions will be discharged. Also due to alternative reaction, soluble oxygen will be converted to OH^- , as

$$O_2 + H_2O + 4e^- \rightarrow 4OH^-$$

These ions (OH^-) are attracted by the anode where they combine with Fe^{2+} ions to form ferrous hydroxide. This iron near the junction is corroded whereas there is no effect on the copper.

A similar situation arises with single metals. For example, if there is a break in the oxide film on the iron surface, the oxide layer behaves as cathode whereas

the exposed iron becomes anode. this facilitates the corrosion.
this process can be retarded by a number of methods
such as:

1. by alloying the metal with suitable materials is stainless steel (Cr and Ni alloy)
2. by applying a protective coating of paint or of a less corrovable metals Fe, Mg, Al, Cu etc.
3. by the protection of the cathode.

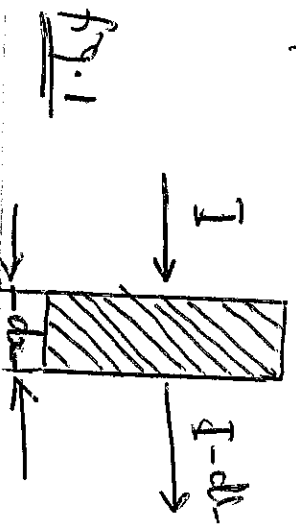
Photochemical Reactions;

Ordinary reactions occur by absorption of heat from outside. The reacting molecules are energised and molecular collisions become effective. These bring about the reaction. The reactions which are caused by heat and in the absence of ~~heat~~ light are called thermal or dark reactions.

On the other hand, some reactions proceed by the absorption of light radiations. These belong to the V and UV regions of the electromagnetic spectrum (2000 to 8000 Å). The reactant molecules absorb photons of light and get excited. These excited molecules then produce the reaction. A reaction which takes place by absorption of the V and UV radiation is called a Photochemical reaction.

Light Absorption

When light is passed through a medium, a part of it is absorbed. It is this absorbed portion of light which causes photochemical reactions. Let a beam of monochromatic light pass through a thickness dx of the medium. The intensity of light reduces from I to $I-dI$.



The intensity of radiation can be defined as the no. of photons that pass across a unit area in unit time.

Let denote the radiations of incident photons by N and the no. of absorbed in thickness dx by dN - the fraction of photons absorbed is then dN/N which is proportional to thickness dx . That is,

$$\frac{dN}{N} = b dx = -\frac{dI}{I}$$

where b is proportionality constant called absorption coefficient.

Let us set $I = I_0$ at $x=0$ and integrate, this gives

$$I = I_0 (-bx)$$

$$\text{or } \ln \left[\frac{I}{I_0} \right] = -bx \quad \text{--- (1)}$$

Lambert first derived equation (1) and it is known as Lambert law. Beer extended this relation to solutions of compound in transparent solvents. The eqn (1) becomes

$$\ln \left[\frac{I}{I_0} \right] = -\epsilon Cx \quad \text{--- (2)}$$

where C = molar concentration; ϵ is a constant characteristic of the solute called molar absorption coefficient. The relation (2) is known as the Lambert-Beer law, this law forms the basis of spectrophotometric methods of chemical analysis.

Laws of Photochemistry

(28)

There are two basic laws governing photochemical reactions.

(a) The Grotthius-Draper Law: When light falls on a cell containing a reaction mixture some light is absorbed and the remaining light is transmitted. Obviously, it is the absorbed component of light that is capable of producing the reaction. The transmitted light is ineffective chemically. In the 19th Century, Grotthius and Draper studied a number of photochemical reactions and eventually concluded that "it is only the absorbed light radiations that are effective in producing a chemical reaction."

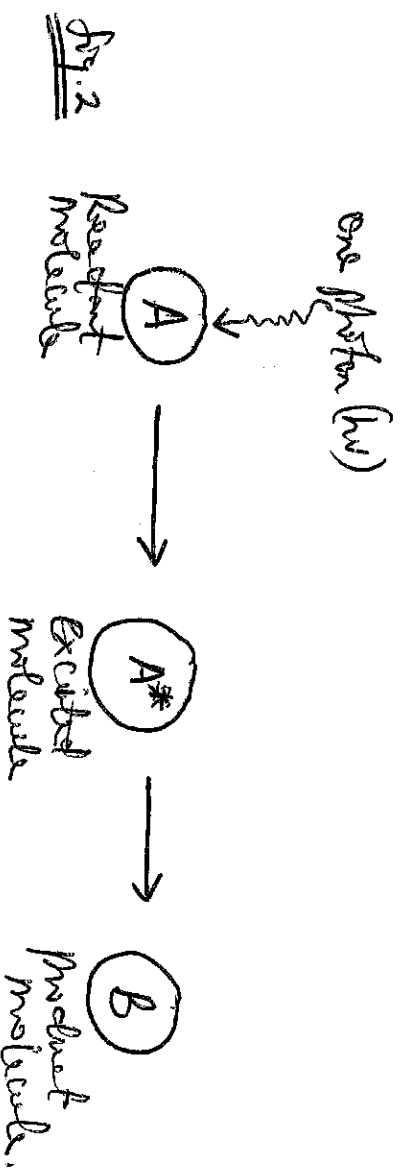
However, it does not mean that the absorption of radiation must necessarily be followed by a chemical reaction. When the conditions are not favourable for the molecules to react, the light energy remains unused. It may be re-emitted as heat or light.

(b) The Stark-Einstein Law of Photochemical Equivalence

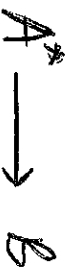
Stark and Einstein (1905) studied the quantitative aspect of photochemical reactions by application of Quantum theory of light. They noted that each molecule taking part in the reaction absorbs only a single quantum or photon of light. The molecule that gains one photon-equivalent energy is activated and enters into reaction. They proposed a basic law of photochemistry which is

named after them and stated that "In a photochemical reaction, each molecule of the reacting substance absorbs a single photon of radiation causing the reaction and is activated to form the products."

The law of photochemical equivalence is illustrated in the figure below:



A molecule 'A' absorbs a photon of radiation and gets excited. The activated molecule (A^*) then decomposes to yield B.



In practice, we use molar quantities. That is, one mole of A absorbs one mole of photon or one einstein. Energy E, the value of E can be calculated by using the expression given below:

$$E = \frac{2.859}{\lambda} \times 10^5 \text{ kcal mol}^{-1}.$$

Quantum Efficiency:

It has been shown that not always a photochemical reaction obeys the Einstein law. The no. of molecules reacted or decomposed is often found to be markedly different from the no. of quanta or photons of radiation absorbed in a given time.

The no. of molecules reacted or formed per photon of light is termed Quantum yield or Quantum efficiency, Φ .

$$\Phi = \frac{\text{No. of molecules reacted or formed}}{\text{No. of photons absorbed.}}$$

For a reaction that obeys strictly the Einstein law, one molecule decomposes per photon, the quantum yield $\Phi = 1$.
 When two or more molecules are decomposed per photon, $\Phi > 1$ and the reaction has a quantum yield. If the no. of molecules decomposed is less than one per photon, the reaction has a low quantum yield.

Calculation of Quantum yield.

By definition, the quantum yield Φ , of photochemical reaction is expressed as:

$$\Phi = \frac{\text{No. of molecules decomposed or formed}}{\text{No. of photons of radiation energy absorbed.}}$$

$$\text{or } \Phi = \frac{\text{No. of molecules decomposed or formed}}{\text{No. of moles of radiation energy absorbed.}}$$

Thus we can calculate quantum yield from -

- (a) the amount of the reactant decomposed in a given time
- (b) the amount of radiation energy absorbed in the same time

The radiation energy is absorbed by a chemical system as photons. Therefore we should know the energy associated with a photon or a mole of photons.

the energy of photons; einstein

the energy of a photon (or quantum), E , is given by

$$E = h\nu = \frac{hc}{\lambda} \quad \text{--- (1)}$$

where h = Planck's constant (6.624×10^{-27})

ν = frequency of radiation

λ = wavelength of radiation (cm, energy in ergs).

c = velocity of light (3×10^{10} cm sec⁻¹)

The energy, E , of an Avogadro N_0 (N) of photons is referred to as one einstein. i.e.

$$E = \frac{Nhc}{\lambda} \quad \text{--- (2)}$$

Substituting the values of N ($= 6.02 \times 10^{23}$), h and c in (2)

$$E = \frac{1.196 \times 10^8}{\lambda} \text{ erg mol}^{-1}.$$

If λ is expressed in Å units ($1 \text{ Å} = 10^{-8} \text{ cm}$).

$$E = \frac{1.196 \times 10^{16}}{\lambda} \text{ erg mol}^{-1}. \quad \text{--- (3)}$$

Ex-Ex 1. cal = 4.184×10^7 erg, energy in calories would be (30)

$$E = \frac{1.196 \times 10^{16}}{\lambda \times 4.184 \times 10^7} \text{ --- (4)}$$

$$= \frac{2.859}{\lambda} \times 10^8 \text{ cal mol}^{-1}.$$

$$\text{or } E = \frac{2.859}{\lambda} \times 10^5 \text{ Kcal mol}^{-1} \text{ --- (5)}$$

It is evident from (3) that the numerical value of Einstein varies inversely as the wavelength of radiation. The higher the wavelength, the smaller will be the energy per Einstein.

Example 1: Cal. the energy associated with (a) one photon,

(b) One Einstein of radiation of wavelength $8000 \text{ \AA}$.
($h = 6.62 \times 10^{-27}$ erg-sec $\times 3 \times 10^{10} \text{ cm sec}^{-1}$).

Solution:

$$\text{(a) Energy of a photon} = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-27} \times 3 \times 10^{10}}{8000 \times 10^{-8}}.$$

$$= \frac{6.62 \times 3}{8.0} \times 10^{-12} \text{ erg}$$

$$= 2.4825 \times 10^{-12} \text{ erg.}$$

$$\text{(b) Energy per Einstein} = \frac{Nhc}{\lambda} = \frac{6.02 \times 10^{23} \times 6.62 \times 10^{-27} \times 3 \times 10^{10}}{8000 \times 10^{-8}}$$

$$= \frac{6.02 \times 6.62 \times 3}{8.0} \times 10^9 \text{ erg}$$

$$= 1.4945 \times 10^{12} \text{ erg.}$$

Example 2. When a substance A was exposed to light, 0.002 mole of it reacted in 20 minutes and 4 seconds.

In the same time A absorbed 2.0×10^6 photons of light per second. Cal. the quantum yield of the reaction. ($N = 6.02 \times 10^{23}$)
Solution:

NO_2 molecules of a reaction = $0.002 \times N = 0.002 \times 6.02 \times 10^{23}$
 NO_2 photons absorbed per second = 2.0×10^6
 NO_2 photons absorbed in 20 min 4 s = $2.0 \times 10^6 \times 1204$

$$\text{Quantum yield } \phi = \frac{\text{NO}_2 \text{ molecules reacted}}{\text{NO}_2 \text{ photons absorbed}}$$

$$= \frac{0.002 \times 6.02 \times 10^{23}}{2.0 \times 10^6 \times 1204}$$

Example 3: When irradiated with light of $5000 \text{ \AA}$, 1×10^{-4} mole of a substance is decomposed.

Photons are absorbed during the reaction. How many efficiency is 10.00. ($N = 6.02 \times 10^{23}$).
Solution.

Quantum efficiency of the reaction = 10.00
 NO_2 moles decomposed = 1×10^{-4}
 NO_2 molecules decomposed = $1 \times 10^{-4} \times 6.02 \times 10^{23}$

$$\therefore \phi = \frac{\text{NO}_2 \text{ molecules decomposed}}{\text{NO}_2 \text{ photons absorbed,}}$$

$$= \frac{6.02 \times 10^{19}}{10}$$

NO_2 photons absorbed,

$$\Rightarrow \text{NO}_2 \text{ photons absorbed} = \frac{6.02 \times 10^{19}}{10}$$

$$= 6.02 \times 10^{18}$$

Example 4. When propionaldehyde is irradiated with light of $\lambda = 3020 \text{ \AA}$, it is decomposed to form CO.



The quantum yield for the reaction is 0.54. The light energy absorbed is 15000 erg/m² in a given time. Find the amount of CO formed in moles in the same time.

Solution

from the equation (3), we have

$$\text{One einstein } (E) = \frac{1.196 \times 10^{16}}{\lambda} \text{ erg/m}^2.$$

$$\text{When } \lambda = 3020 \text{ \AA}, \text{ one einstein} = \frac{1.196 \times 10^{16}}{3020} \text{ erg/m}^2.$$

$$\text{or } 15000 \text{ erg/m}^2 \text{ of energy} = 15000 \times 3020$$

$$\text{But } \phi = \frac{\text{No. of moles of CO formed}}{1.196 \times 10^{16}} = 3.78 \times 10^{-9}$$

$$\text{Hence the amount of CO formed} = \frac{\text{No. of einstein absorbed}}{0.54} = 0.54$$

$$= 0.54 \times 3.78 \times 10^{-9}$$

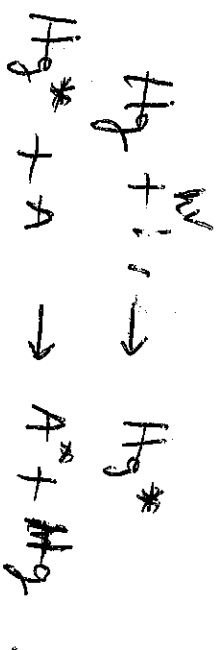
$$= 2.04 \times 10^{-9} \text{ moles}$$

PHOTOSENSITIZED REACTIONS

In many photochemical reactions, the reactant molecule does not absorb the radiation required for the reaction. Hence the reaction is not possible. In such cases, the reaction may still occur if a foreign species such as Mercury Vapor is present. The mercury atoms absorb the incident radiation and subsequently transfer its energy to the reactant molecule which is activated. Thus the reaction occurs. A species which can both absorb and

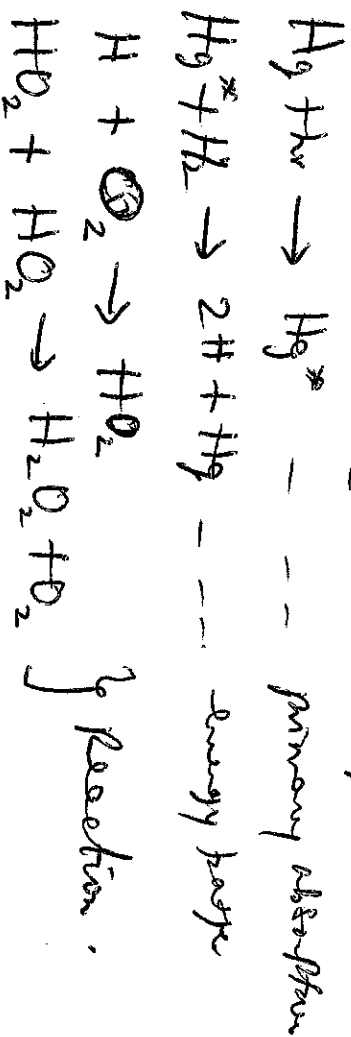
transfer radiant energy for activation of the reactant molecule, is called a photosensitized reaction. The reaction so caused is called a photosensitized reaction.

The role of Mercury Vapor is that of a go-between. The mercury atom absorbs the incident radiation and is excited. The excited atom collides with a reactant molecule (A) and transfers to it the excitation energy. This energy is enough to activate the molecule (A). The mercury atom returns to the original unexcited state.



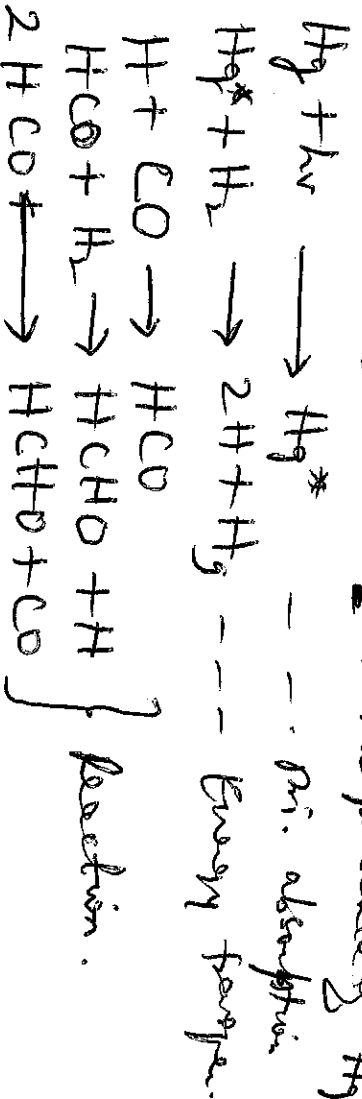
Examples of Photosensitized Reactions:

(a) Reaction b/w H_2 and O_2 molecule is photosensitized by Hg.



Hydrogen peroxide may decompose to form water, H_2O .

(b) Reaction b/w H_2 and CO_2 in the presence of Hg vapor.



PHOTOPHYSICAL PROCESSES

If the absorbed radiation is not used to cause chemical change, it is re-emitted as light of longer wavelength. The two such photophysical processes which can occur are:

③ Fluorescence: Certain molecules (or atoms) when exposed to light radiation of short λ (high frequency), emit light of longer λ . The process is called fluorescence and the substance that exhibits fluorescence is called fluorescent substance. Fluorescence stops as soon as the incident radiation is cut off.

For a solution of quinine sulphate on exposure to V light, exhibit fluorescence. Also a solution of Chlorophyll in ether shows blue-red fluorescence.

from the above, the molecules in solution absorb high energy radiation and excited to higher energy state. Then emits excess energy through several transitions to the ground state. Thus, the excited molecule emits light of longer frequency. The color of fluorescence depends on the λ of light emitted.

④ Phosphorescence: When a substance absorbs radiation of high frequency and emit light when after the incident radiation is cut off, the process is called phosphorescence, and the substance is called phosphorescent substance. It is chiefly caused by UV and V light.

For example, sulphates of Calcium, barium and strontium exhibit phosphorescence. Also, fluorescein in boric acid shows

Phosphorescence in the blue region at $5700 \text{ \AA}$.

As in fluorescence, a molecule absorbs light radiation and gets excited. While returning to the ground state, emits light energy of longer λ . In doing so the excited molecule passes from one series of electronic state to another and gets trapped. This shows the emission of light which persists even after the removal of light source. Thus phosphorescence could be observed as delayed fluorescence.



Chemiluminescence: The emission of light as a result of chemical action is called chemiluminescence. The reaction is called a chemiluminescent reaction. Such a reaction is the reverse of a photochemical reaction which proceeds by absorption of light. The light emitted in chemiluminescent reaction is also called 'cold light' because it is produced at ordinary temperature.

For example, the glow of fireflies due to the aerial oxidation of luciferin (a protein) in the presence of enzyme luciferase. Also, the oxidation of 5-aminophthalic ~~acid~~ cyclic hydroxide (luminol) by hydrogen peroxide in alkaline solution, producing bright green light.

Here, the energy released in the reaction makes the product molecule electronically excited. The excited molecule then gives up its excess energy as visible light while reverting to ground state.

IONIC CONDUCTA

> ionic strength is a measure of the total concentration of ions in a solution. It is the quantity representing the strength of electric field in a solution, equal to the sum of the molalities of each type of ion present multiplied by the square of their charges

> Mos to Steady State Above!

Is a steady state reached by a reacting chemical system when light has been absorbed by at least one of the components, at this state. It can also be defined as a reversible photochemical reaction of photochemical steady state as the equilibrium chemical composition within a system and of electromagnetic irradiation. It is a property of particular importance in photochemical experiments, often used as a measure of their practical efficiency and usually quoted as a ratio percentage.

Effectiveness of light on chemical reactions

light is a source of energy and it can affect the rate of chemical reaction in a number of ways by considerably increasing some & decreasing others. For instance, the reaction between hydrogen and chlorine.

In the presence of light, chlorine molecules are energised and break up into free radicals. The free radical atom called free radicals. The free radical then combine to form hydrochloric acid with hydrogen molecules.

The influence of light on the reaction is described in the reaction shown at ordinary pressure in the following; daylight, darkness.

Reaction:

- In ~~darkness~~ the reaction is slow or almost not occurring.
- In ~~daylight~~ daylight, it is explosive the effect is similar to light in the reaction of alkanes as methane and halogen such as chlorine. Light also influences the photochemical reaction of starch